Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 4

Implications, Vacuous Truth, and Negation

Covers Chartrand & Zhang, Section 1.3, pp. 25–33

Status of this set

Ungraded practice. Solutions are released with it. Attempt every problem before reading any solution. Fill truth tables in on this page; everything else goes on your own paper.

Part I: How to Practice This Material

One row does all the work

Everything in this section follows from a single fact: $P \Rightarrow Q$ is false in exactly one row, when P is true and Q is false. Every other result is a consequence.

Three things to have automatic

1. $P \Rightarrow Q \equiv (\sim P) \vee Q$ (Thm 1.48). When an implication is awkward, rewrite it this way and use the laws you already have.
2. $\sim(P \Rightarrow Q) \equiv P \wedge (\sim Q)$ (Thm 1.50). The negation of an implication is a *conjunction*, never another implication.
3. **A false hypothesis makes the implication true.** Check the hypothesis first — it often settles the question with no further work.

Three mistakes to expect

1. **Negating an implication into an implication.** The negation of "if P then Q" is "P and not Q."
2. **Insisting a true implication needs a real connection.** It does not. The definition looks only at truth values.
3. **Calling a vacuously true statement false or meaningless.** If nothing satisfies the hypothesis, the implication is true.

Part II: Practice Problems

Section A — Truth values

1. Determine whether each implication is true or false, and give a one-line reason.

(a) If 9 is a perfect square, then 9 is odd.

(b) If 9 is a perfect square, then 9 is even.

(c) If 8 is a perfect square, then 8 is odd.

(d) If $\pi > 4$, then $\pi > 3$.

2. Construct a truth table for $(P \Rightarrow Q) \wedge (\sim Q)$, then state in one sentence when it is true.

P	Q	$P \Rightarrow Q$	$\sim Q$	$(P \Rightarrow Q) \wedge (\sim Q)$
T	T			
T	F			
F	T			
F	F			

Section B — Vacuous truth

3. Decide whether each statement is true, and say why.

(a) Every integer n with $n^2 < 0$ is prime.

(b) Every element of the empty list is greater than 100.

(c) Every prime number greater than 2 is odd.

Section C — Negating implications

4. Write the negation of each. Phrase it the way a person would say it.

(a) If the connection drops, the transaction is rolled back.

(b) If n is divisible by 6, then n is divisible by 4.

(c) If a student passes the final, then the student passes the course.

(d) If the array is empty, then the function returns zero.

Section D — Implications with open sentences

5. For a real number x, let $P(x) : x = 3$ and $Q(x) : x^2 = 9$.

(a) State $P(x) \Rightarrow Q(x)$ in words.

(b) Is $P(a) \Rightarrow Q(a)$ true for *every* real number a? Justify.

(c) Is $Q(a) \Rightarrow P(a)$ true for every real a? If not, give a value of a where it fails.

Section E — Specifications

6. A system requirement reads:

> *If a request is authenticated, then it is logged.*

(a) Which single situation would show the system violates this requirement?

(b) An auditor finds an unauthenticated request that was logged. Does this violate the requirement?

(c) The system logs nothing at all for a week because it received no requests. Does this violate the requirement?

Section F — Writing

7. In five to eight complete sentences, explain to a classmate why "If $2 + 2 = 5$, then the moon is made of cheese" is a *true* statement. Your explanation should address why this feels wrong and why the definition is set up this way anyway. Use the contract idea from class.

Section G — Challenge (optional)

8. Use Theorem 1.48 and the laws from Section 1.2 to show that $P \vee Q \equiv (\sim P) \Rightarrow Q$, without building a truth table.